

Anton Achhammer

Curriculum Vitae

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Education

- Mar. 2021 – Mar. 2023 **M.Sc in Computer Science**, *Technical University of Applied Sciences*, Regensburg, Germany
Main courses: distributed computing, artificial intelligence, modern theoretical computer science, modern database concepts, modern software techniques.
Skills: NoSQL, Python, JavaScript, $\LaTeX$.
Project contributions: “Waste Management: Route optimization for waste disposal” - recording of garbage can filling levels and optimization of the route, “Car charging station app”.
Student research paper: “Vertical Farming: How can AI and ML optimize plant growth”.
Thesis: “Design and implementation of a health score for calves” using sensor technology and AI.
- Oct. 2017 – Mar. 2021 **B.Sc in Computer Science**, *Technical University of Applied Sciences*, Regensburg, Germany
Main courses: software engineering, innovation management, business administration, IoT, design thinking.
Skills: C, Java, HTML, CSS, JavaScript, Matlab, SQL, Assembler, $\LaTeX$.
Thesis: “Conception and implementation of a visualization of automated warehouse systems”.
- Sep. 2009 – Jun. 2017 **A levels**, *Albertus Magnus Gymnasium*, Regensburg, Germany
Main subjects: mathematics, computer science, english.
Skills: Java, HTML, Assembler, SQL.
Voluntary activities: member of the school’s medical service, member of the school’s technology team, tutor in the prevention projects “Netzgänger” and “AMG against cyberbullying”, mentoring tutor.

Work Experience

- Mar. 2024 – today **Lecturer**, *Technical University of Applied Sciences*, Regensburg, Germany
Lectures and supervision of project work for students on the bachelor’s degree program in Renewable Energy Technology and Energy Efficiency on the topic of “Sustainable energy system modeling using PyPSA”.
- May 2023 – today **Research Assistant**, *Institute for Energy Networks and Energy Storage (FENES)*, Regensburg, Germany
Modelling and development of low carbon energy systems with focus on sector coupling and energy storage, Co-Development of the open source energy system model “PyPSA Earth”, Deputy project manager in the “H2Global meets Africa” project, supervision of student assistants and thesis projects, budget planning with CEUS.
Skills: Python, PyPSA, $\LaTeX$, CEUS.
- Jan. 2025 – Mar. 2025 **Guest Researcher**, *University of Oxford*, Oxford, UK
Development of open energy system models
Skills: Python, PyPSA.

- Sep. 2024 – Oct. 2024 **Guest Researcher**, *Council for Scientific and Industrial Research (CSIR)*, Pretoria, South Africa
Research on open energy system modelling
Skills: Python, PyPSA.
- Apr. 2022 – Sep. 2022 **Thesis Student**, *Futuro Farming GmbH*, Regensburg, Germany
Reading of movement data via sensors - prediction of diseases via AI.
Skills: Python, AWS, Pandas.
- Aug. 2019 – Feb. 2023 **Intern, Working Student, Thesis Student**, *TGW Software Services GmbH*, Regensburg, Germany
Working in the fields of software development and project management.
Agile software development with Scrum and JIRA, effort estimates, specification, project management, software tests, knowledge transfer via Confluence, bug fixing.
Skills: Java, SQL, L^AT_EX.
- Jul. 2014 **Intern**, *Continental Automotive*, Regensburg, Germany
Skills: C#, SQL.

Scholarships and Awards

- Jun. 2024 – today **First place at several Science Slams**
Science slam on “Modeling socially and climate-friendly hydrogen partnerships”.
First place in:
 - Regensburg (Regensburger Nachhaltigkeitswoche 06/2024)
 - Berlin (ScienceSlam.de - Zeiss Großraumplanetarium 12/2024)
 - Tannesberg (Förderverein Haus der Biodiversität e.V. 04/2026)
- Jul. 2026 **MLP Scholarship - International Category**
Awarded for outstanding academic achievements, educational work and intercultural commitment. Selected through a multi-stage national selection process (25 scholars nationwide per year).
- Sep. 2025 **BayFor WKS Bavaria-Africa Mobility Grant**
Scholarship for research trip to East Africa
- Jun. 2025 **Sparkassen Sustainability Award**
Award for particularly sustainable behavior, granted for my work on educational materials for open-source energy models
- Jan. 2025 **Erasmus+ scholarship**
Scholarship granted for my research visit at the University of Oxford
- Oct. 2024 **Finalist West German Science Slam**
Science slam on “Modeling socially and climate-friendly hydrogen partnerships”
- Jul. 2024 **Future energy labs (FEL) scholarship of the German Energy Agency (dena)**
Scholarship granted for my research trip to Southern Africa

Publications & Reports

- 2026 Schumm, L., **Achhammer, A.**, Sterner, M., *The global, country-wise potential of green hydrogen in renewable-dominated electricity systems*, in *International Journal of Hydrogen Energy: Special issue on SDEWES-2025*, doi: 10.1016/j.ijhydene.2026.156641.
- 2026 **Achhammer, A.**, Fioriti, D., Patonia, A., Sterner, M., *The impact of hydrogen underground storage on export dynamics and system resilience: A GIS-based integration of salt caverns into PyPSA-Earth*, in *Energy: Special issue on Smart Energy Systems SESAAU2025*, doi: 10.1016/j.energy.2026.141729.
- 2025 Schamel, M., **Achhammer, A.**, Schumm, L., Sterner, M., *Harvesting Sustainability: Cost-competitiveness of Green Fertilizer Value Chains in Western Africa*, in *IEEE Xplore: Proceedings of the IEEE PES/IAS PowerAfrica Conference 2025*, doi: 10.1109/PowerAfrica65840.2025.11289137.
- 2025 Uhl, J., Schamel, M., **Achhammer, A.**, Sterner, M., *Sustainable Steel Production in the Desert: Economic and Technical Assessment of a Hydrogen-Powered Steel Plant in Mauritania*, in *IEEE Xplore: Proceedings of the IEEE PES/IAS PowerAfrica Conference 2025*, doi: 10.1109/PowerAfrica65840.2025.11289120.

Third-Party Funding & Research Projects

- Nov. 2026 – today **GREEN-HET**, Erasmus+ Capacity Building in Higher Education (CBHE), Developing a Master's program in hydrogen-enabling technologies in Sri Lanka (total budget approx. EUR 800,000).
- May 2026 – today **H2Map-Educate**, Collaborative project with Namibia and South Africa: Linking digital infrastructure mapping with socio-ecological risk assessments and capacity building. Funded by BMFTR / PtJ (total budget approx. EUR 800,000).
- Jul. 2026 – today **Open vhb: Modeling sustainable energy systems**, Grant by the Virtuelle Hochschule Bayern (vhb) for the development of an open-access online course (Open vhb) about energy system modeling with PyPSA (total budget approx. EUR 20,000).
- Dec. 2025 – today **HySee: Hydrogen for Southeast Europe**, Supporting renewable hydrogen market development in Bulgaria and Romania and building a South-East European hydrogen corridor towards Central Europe (total budget approx. EUR 796,000).
- Sep. 2025– Feb. 2026 **Classic vhb: Modeling sustainable energy systems**, Grant by the Virtuelle Hochschule Bayern (vhb) for a bilingual university-level digital course about energy system modeling with PyPSA (total budget approx. EUR 90,000).
- Jan. 2023– Jun. 2026 **H2Global meets Africa**, Collaborative research project on techno-economic energy system analysis and PtX value chain modeling (PyPSA-Earth); funded by BMBF (total budget approx. EUR 4.2 million).

Teaching & Lecturing Appointments

- since WT 24/25 **Energy System Modeling**, Lecture, Technical University of Applied Sciences (OTH) Regensburg.
- ST 24 **Project Term Paper in Renewable Energy Technology and Energy Efficiency**, Project course and seminar supervision, Technical University of Applied Sciences (OTH) Regensburg.

Skills

Programming Languages	Python (excellent), Java (excellent), SQL (excellent), C# (very good), C (very good), JavaScript (good) Profound knowledge in energy system modelling languages
Languages	Deutsch (native), Englisch (C1), Latein (intermediate certificate) Fluent in German and English
Applications	Microsoft Office (excellent), L^AT_EX(very good) Excellent knowledge in Microsoft Office applications and L ^A T _E X for publications

Voluntary Activities

Oct. 2025 – today	Faculty Council , <i>Technical University of Applied Sciences</i> , Regensburg, Germany Representative of the scientific employees.
Oct. 2023 – today	Volunteer , <i>Scientists for Future</i> , Regensburg, Germany Support of the local group in Regensburg.
Oct. 2023 – today	Volunteer , <i>Viva con Agua</i> , Regensburg, Germany Support of various charity and educational campaigns of the local group in Regensburg.
Dec. 2011 – today	Firefighter , <i>Freiwillige Feuerwehr Pettendorf</i> , Pettendorf, Germany
Dec. 2022 – today	IT Representative , <i>Freiwillige Feuerwehr Pettendorf</i> , Pettendorf, Germany
Nov. 2020 – today	Webmaster , <i>FC Pielenhofen-Adlersberg</i> , Pettendorf, Germany
Oct. 2023 – Jul. 2025	Volunteer , <i>Ratisbona Coding e.V.</i> , Regensburg, Germany Support of educational events.
Nov. 2021 – Mar. 2025	IT Representative , <i>TSV Adlersberg</i> , Pettendorf, Germany
Sep. 2014 – Jul. 2017	School first aid service , <i>Albertus-Magnus-Gymnasium Regensburg</i> , Regensburg, Germany
Jul. 2016	Media tutor in the “AMG against cyberbullying” prevention project , <i>Albertus-Magnus-Gymnasium Regensburg</i> , Regensburg, Germany
Sep. 2013 – Jul. 2016	Member of the technology team , <i>Albertus-Magnus-Gymnasium Regensburg</i> , Regensburg, Germany
Mar. 2015 – Jul. 2016	Media tutor in the “Netzgänger” prevention project , <i>Albertus-Magnus-Gymnasium Regensburg</i> , Regensburg, Germany
Sep. 2013 – Jul. 2015	Mentoring tutor , <i>Albertus-Magnus-Gymnasium Regensburg</i> , Regensburg, Germany